Sizing the put book down when VIX is high cuts its drawdown but not its Sharpe before 2020

Experiment report · L/S Gamma Desk, QUANTT

2026-11-02

Question: On the 35% stress-sized `hedged` put book, does scaling each entry's stress budget by VIX's one-year percentile give a higher Sharpe ratio than the fixed 35% budget, over the full window and in each half, with a less severe maximum drawdown, at 1x and 2x spreads? The scale runs from 1.5x when VIX is at its one-year low to 0.5x at its one-year high.

Method:

- **Pre-registered backtest.** The research switches were committed in a2d5334 and the hypothesis card in e31b598, before any run.
- **Data and window.** The live paper rules (`configs/paper.toml`) are replayed on R2 replay snapshots v1 (config_hash b040760187f0) over the development window 2014-06-02 to 2024-12-31. The hold-out (2025-01-01 to 2026-07-10) was not used.
- **Book (all variants and the reference):** the `hedged` book of `2026-10-22-stress-sized-puts`.
  - An always-on short 25-delta put, with the VRP signal ignored and the flat exit off.
  - Sized at entry so a one-day 20% SPY drop, with IV at max(60%, the put's IV), costs at most 35% of $250k.
  - Hedged daily at 10:15 with a 10-share band. The 1x-credit stop, 50% profit-take, 7-DTE time exit and 1-day cool-off are as live.
- **Switches.** `size_by` and `size_tilt` multiply the stress budget by $1 + \text{tilt} \times (2p - 1)$, where p is the percentile of the prior close among the last 252 completed closes. `size_cap` caps that multiple. This is a rank version of 1/VIX, not a literal ratio.
- **Variants:**
  - `vix_inv` (primary): `size_by = "vix"`, tilt −0.5, so 1.5x at VIX's one-year low and 0.5x at its high.
  - `vix_inv_cap`: the same, capped at 1x (`size_cap = 1.0`), so the book is never above the budget.
  - `rv_inv`: `size_by = "rv_m"` (monthly realized volatility), tilt −0.5, a stand-in for sizing by 1/GARCH.
- **Comparison:** the registered `2026-10-22-stress-sized-puts/hedged` runs, reused and not rerun, at the same spread multiple.
- **Runs and costs:** every variant at 1x and 2x spreads. $0.65 per contract and $0.005 per share, with a $1 minimum per order. Decisions, exits and hedges run once a day at 10:15 ET.

1 Result vs the pass bar

Every check had to hold at 1x and 2x spreads. P&L and drawdown are in USD on $250k of capital. Sharpe is annualised from daily net P&L.

Check (`vix_inv` vs `hedged`)	**1x value**	**1x bar**	**2x value**	**2x bar**	**Result**
Sharpe, full window	0.62	> 0.52	0.48	> 0.43	pass
Sharpe, 2014-06 to 2019-12	0.78	> 0.84	0.61	> 0.72	fail
Sharpe, 2020-01 to 2024-12	0.57	> 0.48	0.42	> 0.38	pass
Max drawdown less severe	$41,489	$< \$70{,}362$	$45,307	$< \$75{,}488$	pass
Round trips (floor 30)	183	≥ 30			met

Verdict: failed. Sizing down when VIX was high raised the full-window Sharpe and cut the worst drawdown by $29k at 1x, but it lowered the Sharpe in 2014-2019 at both spreads. Net P&L fell from $121.5k to $95.4k at 1x and from $101.2k to $74.8k at 2x. The book still makes money.

1.1 Full-window numbers (not part of the bar)

1x spreads

	hedged	**vix_inv**	**vix_inv_cap**	**rv_inv**
Net P&L	$121,482	$95,448	$83,364	$101,878
Sharpe	0.52	0.62	0.59	0.65
Max drawdown	$70,362	$41,489	$38,887	$51,083
ES 5% (daily)	−$3,016	−$2,454	−$2,203	−$2,486
Round trips	182	183	183	187
Contracts per entry, mean / max	25.4 / 100	24.8 / 53	21.2 / 49	25.0 / 55
Commissions, options / hedges	$5,955 / $2,868	$5,877 / $2,866	$5,011 / $2,675	$6,064 / $2,879

2x spreads

	hedged	**vix_inv**	**vix_inv_cap**	**rv_inv**
Net P&L	$101,172	$74,804	$67,415	$93,306
Sharpe	0.43	0.48	0.47	0.59
Max drawdown	$75,488	$45,307	$42,464	$48,190
ES 5% (daily)	−$3,094	−$2,531	−$2,264	−$2,512

No variant passes the 2014-2019 Sharpe check: `vix_inv_cap` has 0.77 and `rv_inv` 0.67, against 0.84 at 1x.

2 Charts

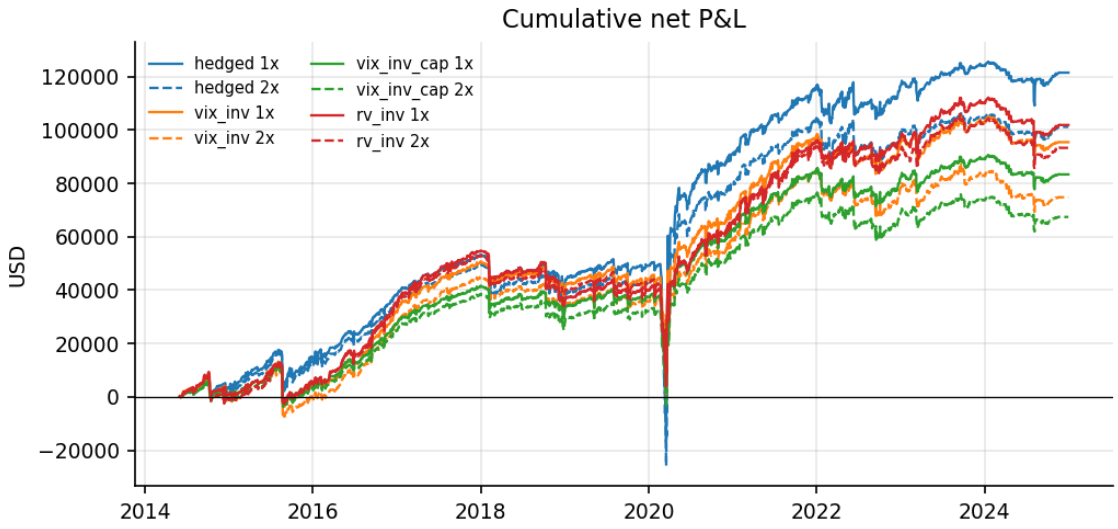


Figure 1: Cumulative net P&L, 1x solid and 2x dashed

All four books hold the same put on most days: 178 of `vix_inv`'s 183 entries fall on the same day as an entry of `hedged`. `vix_inv` falls behind in August 2015 and in early 2020 and does not catch up. It gains back a little in 2016, 2017 and 2021, when VIX was low and it held more contracts, and in 2018, when it was smaller in the February and December spikes.

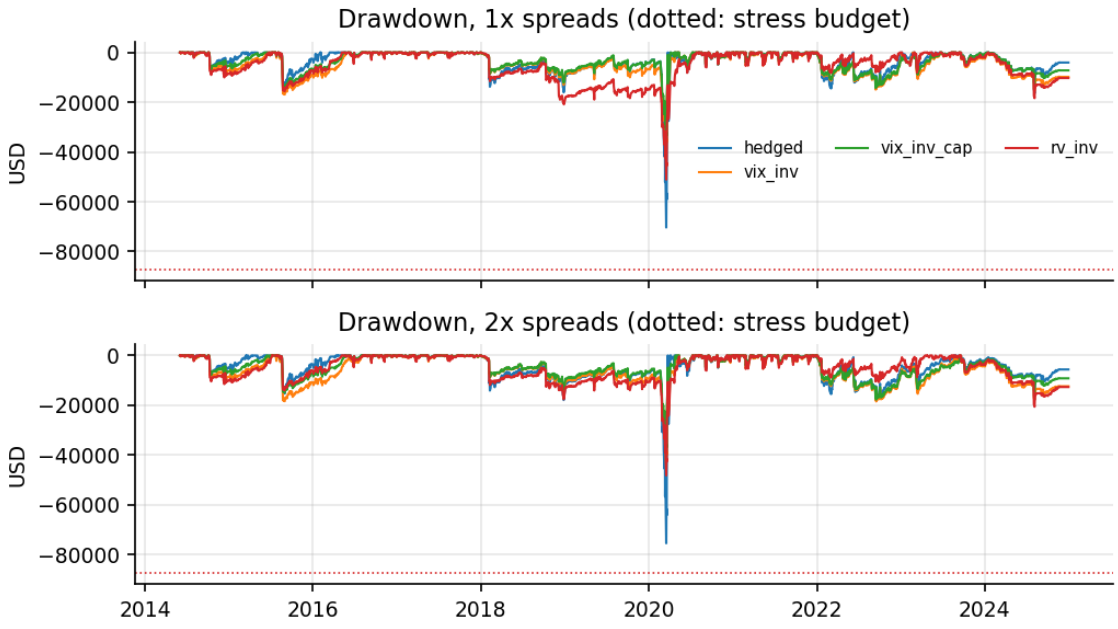


Figure 2: Drawdown at 1x and 2x spreads; dotted line: the $87,500 stress budget

Every book's deepest drawdown falls on 17 March 2020. `vix_inv` loses $41k there against `hedged`'s $70k, because it entered the spike at half size. `rv_inv` also stays underwater from January 2018, and its drawdown

deepens from late 2018 to early 2020. That is path dependence: after a September 2018 profit-take it re-entered on dates the `hedged` book did not. Its seven round trips entered from mid-September 2018 to early January 2019 on such dates lost about $15k in total.

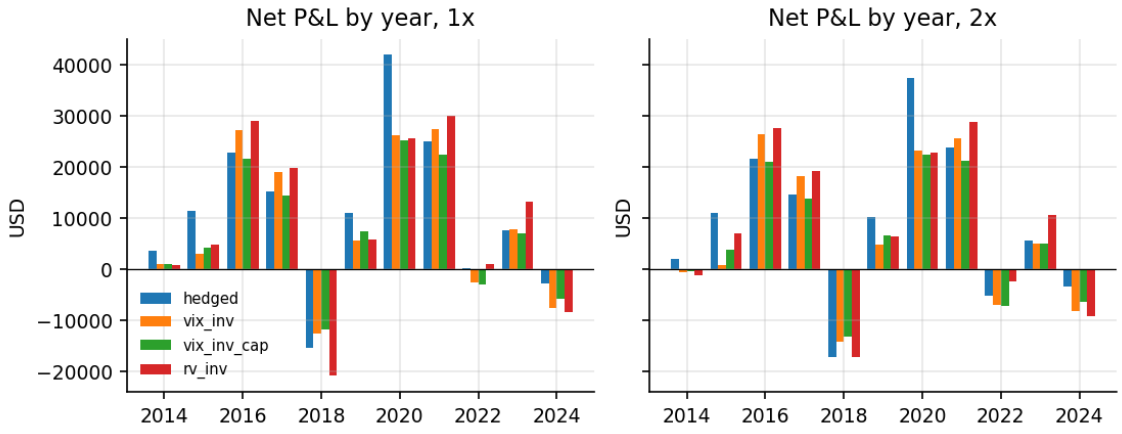


Figure 3: Net P&L by year

At 1x, `vix_inv` makes less than `hedged` in 6 of 11 years. The largest gaps are 2020 (−$15.8k), 2015 (−$8.4k), 2019 (−$5.4k) and 2024 (−$4.8k). It makes more in 2016, 2017, 2018, 2021 and 2023, by $0.3k to $4.4k each.

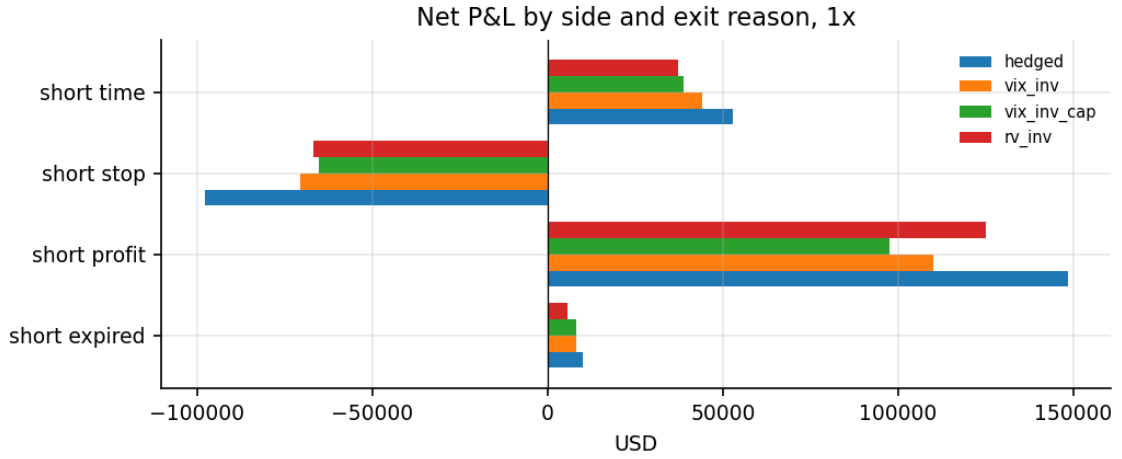


Figure 4: Net P&L by side and exit reason, 1x

`vix_inv` cuts stop losses by $27.2k, but its profit-takes shrink by $38.3k and its time exits by $8.9k. It sold less on stressed days whose puts later paid off.

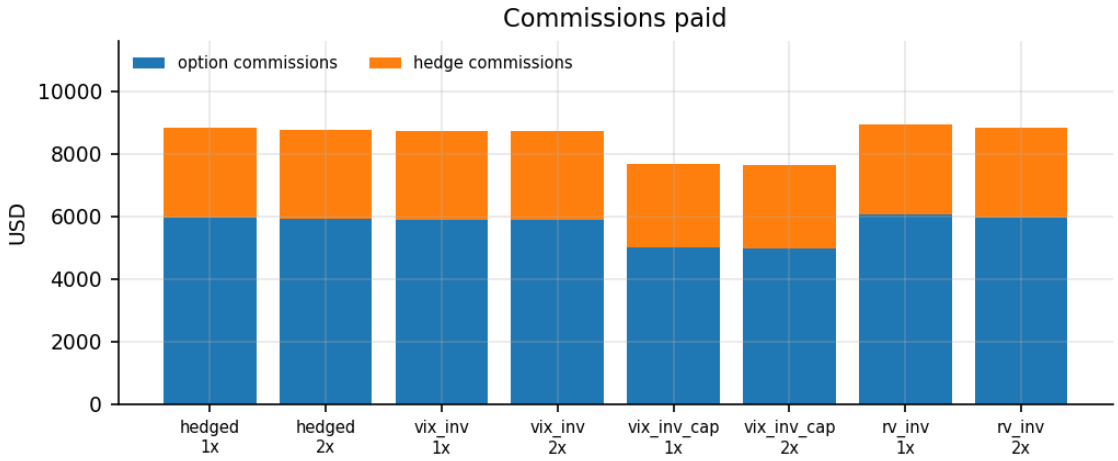


Figure 5: Commissions paid

Commissions barely move (about $8.7k to $8.9k), except for the capped book, which is smaller on average and pays about $1.1k less. Costs don't explain the result.

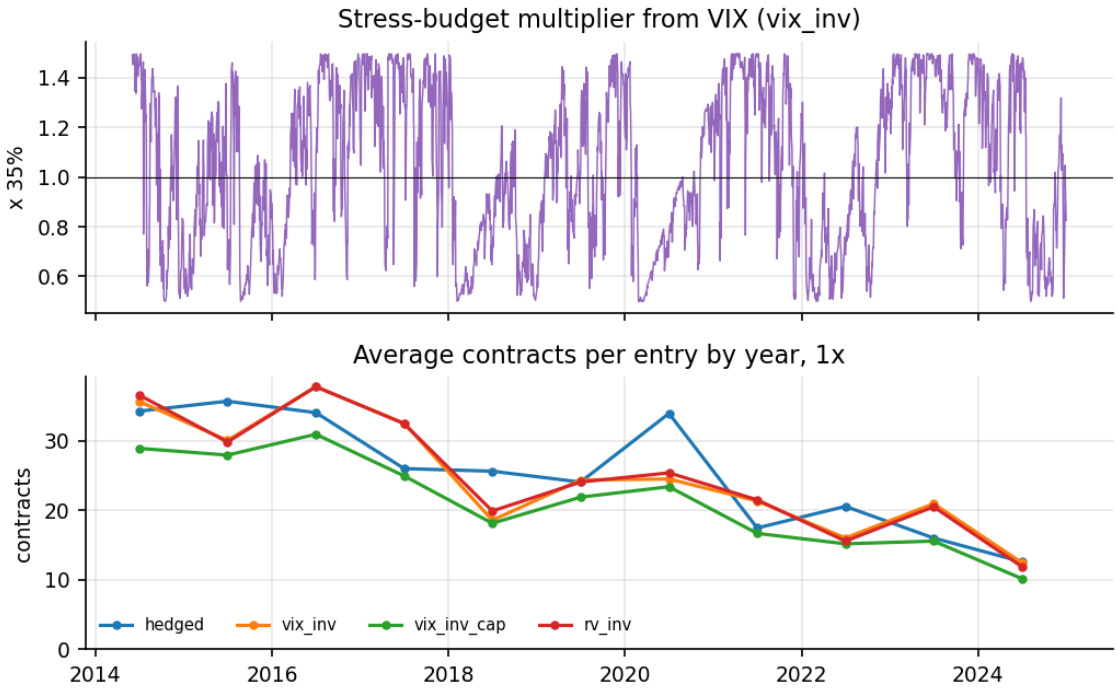


Figure 6: Stress-budget multiplier from VIX's one-year percentile, and average contracts per entry by year

The multiplier swings between its bounds within weeks, because VIX's percentile moves fast after a spike. Average size at entry is almost unchanged (1.03x mean at entry; 49% of entries below 1x). The tilt moves size between years: more in 2016-2017, 2021 and 2023; less in 2015, 2018, 2020 and 2022.

3 Analysis

The Sharpe gain is the March 2020 cut.

- All five entries from 26 February to 17 April 2020 came at 0.50x to 0.61x, because VIX was at or near its one-year high at each prior close.
- The 100- and 71-contract entries of `hedged` became 49 and 37 contracts.
- From 19 February to 30 April 2020, `vix_inv` made \$9.0k against `hedged`'s \$21.7k at 1x, and its drawdown fell from \$70k to \$41k.
- Outside that window, `vix_inv` has a lower Sharpe than `hedged` (0.75 against 0.81 at 1x) and makes \$86.5k against \$99.8k. The same holds for `vix_inv_cap` (0.73) and `rv_inv` (0.78).

Before 2020 the tilt sized the wrong way around shocks.

- A put entered at a calm low VIX is sized up just before a spike. One entered right after a spike is sized down, though those puts are the richest and usually recover.
- In 2015, the entry in mid-August came at 1.14x and was stopped out. The entry the day after the 24 August spike came at 0.5x and made about half what the `hedged` entry made.
- Around the December 2018 selloff the effects cancel: the 14 December entry at 0.65x made less in the year-end rebound (about −\$1.6k), while being smaller in the December fall gained \$2.7k. Over December 2018 and January 2019 combined the books are level.
- In 2014-2019, mean daily P&L fell 11% (\$32.3 against \$36.1) and its volatility only 4%, so the Sharpe fell.

Per-contract P&L is highest at low VIX, but the dollars are at high VIX. In the `hedged` book, mean net P&L per contract by multiplier tercile at entry was:

- \$7.9 in the high-VIX tercile (multiplier about 0.64x);
- \$1.1 in the middle tercile;
- \$21.9 in the low-VIX tercile (about 1.40x).

So the tilt does put more contracts where the reference earned more per contract. But the high-VIX tercile carried the most dollars (\$71.2k of \$113.2k round-trip net), from a few large crash entries, and the tilt halves those.

The result is mostly mechanical: a smaller book in the crash. The yearly differences outside 2020 are a few thousand dollars and change sign, so the evidence of timing skill either way is weak.

4 Verdict

Failed, per the pre-registered bar: Sharpe in 2014-2019 is lower at both 1x and 2x. The tilt behaves like a smaller budget in crashes, as the slope-sizing card found, and it trails the fixed budget outside March 2020. Archive the card and don't run the hold-out. The 15% and 25% budgets and the 1.5×IV stress (`2026-11-02-crash-shock`) are the direct test of cutting crash size.

5 Caveats

- **In-sample:** the book was chosen after its 2014-2024 results, and earlier cards on it (slope sizing, `vvix_reverse`) had already shown that sizing down in stress raises the full-window Sharpe. Only the hold-out could confirm a result.
- **One episode:** the full-window Sharpe and drawdown results rest on five entries in early 2020.
- **Rank, not ratio:** the multiple uses VIX's one-year percentile, not 1/VIX, so it reverses quickly after a spike. A literal 1/VIX ratio might behave differently.
- **Spread model:** fitted on about two weeks of calm 2026 live quotes. The 2x run is the stress test, and large March 2020 fills use the same model as small ones.
- **Data gaps:** R2 has no month-end weekdays, so about 3.3% of scheduled times are skipped.
- **Margin:** not modelled.

6 Trials

6 trials for this experiment (3 variants × 1x and 2x, one run of `experiment.py`), plus the baseline's 2. The `hedged` reference was reused, not rerun. Every row of this run is tagged `-dirty`, and the commit tags differ (3ab834b, af47198, 79b09a1, 3568520). Other cards of the same batch were committing their pre-registrations and writing files in the shared working tree during the run. Each tagged commit contains this card's pre-registration (e31b598) and changes no engine code, `backtest.toml`, or this card's config or files. The pass bar was not changed.